

Project Design Phase Proposed Solution Template

Date	20 February 2026
Team ID	LTVIP2026TMIDS88068
Project Name	Online Payments Fraud Detection using Machine learning
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	The increase in online payment transactions has led to a rise in fraudulent activities and unauthorized transactions. Traditional systems fail to detect fraud in real time, causing financial loss and reduced customer trust.
2.	Idea / Solution description	Develop a Machine Learning-based Online Payment Fraud Detection System that analyzes transaction details such as amount, transaction type, and account balance to automatically classify transactions as fraud or legitimate using trained ML models.
3.	Novelty / Uniqueness	The system uses machine learning algorithms to identify hidden fraud patterns instead of traditional rule-based detection. It provides automated real-time prediction and improves accuracy over time through data learning.

4.	Social Impact / Customer Satisfaction	Enhances digital payment security, protects users from financial loss, increases customer confidence in online transactions, and promotes safe digital financial services.
5.	Business Model (Revenue Model)	The system can be integrated into banking applications or payment gateways as a fraud detection service. Revenue can be generated through subscription-based security services, API integration fees, or enterprise licensing.
6.	Scalability of the Solution	The solution can handle large volumes of transaction data and can be scaled for banks, e-commerce platforms, and digital payment systems using cloud deployment and real-time processing